[1] Ranjana
[2] Dr. Sarvottam Dixit
[3] Dr. Pooja Tripathi

IoT Network Intrusion Detection and Classification using Explainable (XAI) Machine Learning Algorithms

Journal of Electrical Systems

Abstract: - Artificial intelligence (AI) has grown significantly, but because it is a "black box," it is difficult to build enough trust. As a result, it is rarely used as a stand-alone device in high-risk Internet of Things applications including financial, medical, and critical industrial infrastructures. Explainable AI (XAI) has been developed to address this issue. But creating suitably quick and precise XAI remains difficult.

XAI can be used to provide explainability of machine learning techniques. As explainability has become a fundamental issue with the newest ML techniques, like ensembles or Deep Neural Networks, XAI has gained significant traction recently as accountability, transparency, trust, so model explainability is gaining priority. More precisely, [1] have carried out preliminary research into creating a network analysis for mischievous behaviours for explainable artificial intelligence frameworks. When machine learning models become more accurate, their complexity rises and interpretability falls as a result.

In this research Deep Neural Networks have been created to detect intrusion at network along with an explanation module for building trust in the building model through AI algorithms. Using the dataset on intrusion detection system, we successfully increased model transparency by applying the existing XAI algorithms, which provide explanations on individual predictions generated, such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), Contrastive Explanations Method (CEM), Protidic, and Boolean Decision Rules via Column Generation (BRCG).

Keywords: IoT, Intrusion Detection, Cybersecurity, Cyber Attacks, Explainable AI, Trust, Ethics.

I. Introduction

Modern and sophisticated machine learning techniques and models have important uses in improving IoT network security. Machine learning algorithms are designed to learn from network traffic features and assist cyber security experts in making important decisions related to threat identification. However, they might resort to more conventional methods, which might not be as effective but provide greater explainability or human trust. Gaining an understanding of the parameters of the networks for an ML system for Intrusion Detection is often practically unattainable. This could further erode confidence in a particular forecast made. Cyber security professionals will be in a better position to trust predictions of the classifiers, especially when it comes to understand how they act in specific circumstances, provided they have an intelligible analysis of the reasoning for each prediction. In order to contribute to the increase of general confidence. Explainable AI (XAI) provides a range of feature importance or explanation tools for trained machine learning models.

II. Literature Review

Zhou et al. (2020) developed a DNN-based IDS that achieved high detection accuracy for various types of attacks on IoT networks. The model utilized a multi-layer architecture to capture intricate patterns and relationships in the data. Alrawashdeh and Purdy (2016) proposed a deep learning approach for network intrusion detection, highlighting the superiority of DNNs over traditional machine learning models in terms of accuracy and robustness. Afanasev et al. (2021) demonstrated the application of XAI techniques to a DNN-based IDS for a smart home IoT network. They used SHAP to identify key features contributing to the detection of specific attack types, improving the system's interpretability. Yang et al. (2022) applied LIME to interpret a DNN model used for detecting Distributed Denial of Service (DDoS) attacks in an industrial IoT environment. Their approach provided actionable insights into the model's decision-making process, enhancing its reliability. Ahmed, A., &

[1, 2] Research Scholar, Mewar University, Gangarar, Rajasthan, India

[2] Professor, Computer Science, Mewar University, Gangarar, Rajasthan, India

[3] Professor, Information Technology, Inderprastha Engineering College, Ghaziabad, Uttar Pradesh, India

Copyright © JES 2024 on-line : journal.esrgroups.org

Mahmood, A. N. (2021) reviews the state of XAI in cybersecurity, focusing on its application in intrusion detection and providing a roadmap for future research.

Chaudhary, R., Saini, H., & Awasthi, L. K. (2021) proposed an IDS for IoT networks leveraging XAI techniques to improve the transparency and interpretability of DNN models. Li, Y., Zhu, Y., & Zheng, W. (2022), presents an XAI framework for DNN models in IoT, emphasizing real-time anomaly detection and interpretability. Gomes, R. P., Silva, A. G., & Pereira, R. (2022), uses Layer-Wise Relevance Propagation (LRP) to enhance the interpretability of DNN models in IoT intrusion detection. Tran, D. H., Nguyen, H. M., & Pham, X. Q. (2023), present a comprehensive XAI framework for interpreting DNN-based IDS in IoT networks, emphasizing its practical applications and benefits. Jain, R., Gupta, V., & Kumar, R. (2023), discusses the integration of various XAI methods with DNN-based IDS to improve transparency and trust in IoT security systems. Sharma, K., & Bhargava, B. (2023), A survey of current XAI techniques applied to ML models for detection of intrusion in IoT, providing a comprehensive survey of methodologies and their effectiveness. Wang, S., & Hu, X. (2024), explores the challenges and opportunities in developing explainable deep learning models for IoT security, with a focus on intrusion detection systems. Zhao, Z., & Li, J. (2024), presents a comprehensive XAI framework for enhancing the interpretability of DNN-based intrusion detection systems in IoT networks.

Kim, Y., & Park, J. (2021), explores deep learning-based anomaly detection in IoT networks and applies XAI techniques to improve model interpretability and trustworthiness. Patel, S., & Gupta, R. (2021), investigates how XAI can be used to enhance the interpretability of security models in IoT networks, focusing on DNN applications. Nguyen, D. T., & Hoang, N. T. (2022) proposed a hybrid model combining deep learning and rule-based techniques for intrusion detection, providing explanations for model. Petersen, R., & Schmidt, T. (2023) focuses on the challenges and solutions for implementing XAI in DNN-based intrusion detection systems for IoT networks

Feng, H., & Wang, Z. (2023) examines how deep learning and XAI can be combined to improve the interpretability and effectiveness of intrusion detection systems in IoT environments. Gomez, C., & Martinez, F. (2024) explore various explainable deep learning approaches to enhance the security of IoT networks, with a focus on practical applications and deployment. Rahman, M., & Islam, S. (2024) discusses the development of AI-driven explainable intrusion detection systems specifically designed for smart IoT environments. Hassan, R., & Ali, M. (2024) integrates XAI techniques with deep learning models to enhance the security and interpretability of IoT networks, providing detailed case studies and applications. Smith, J., & Jones, L. (2024) review of XAI techniques for intrusion detection in IoT networks, highlighting current challenges and potential future research directions

III. TECHNIQUES

The UNSW-NB15 Catalog

An Internet of Things (IoT)-based network traffic data collection called UNSW-NB15 has distinct categories for both legitimate network traffic and malicious botnet attack behaviors (via attack type classification including Shellcode, Worms Analysis, Backdoors, Generic, Reconnaissance, Fuzzers, DoS, Exploits). The UNSW-NB15 testbed configuration dataset and feature creation processes are depicted in Figure 1which was used in the Cyber Range Lab of the Australian Center for Cybersecurity.

The UNSW-NB15 dataset was prepared by splitting it in to two parts as training set and testing set, with the goal being to observe the behavious of the traffic as attack, and normal behavior. The dataset contains 39 attributes that are of a numerical type. In Figure 2, the class of each attack within the data subsets is displayed together with its characteristics and value distribution, with "0" and “1” represents normal behavior and assault behavior. It is evident that the dataset is suitably balanced with respect to the activity behavior binary response variable.

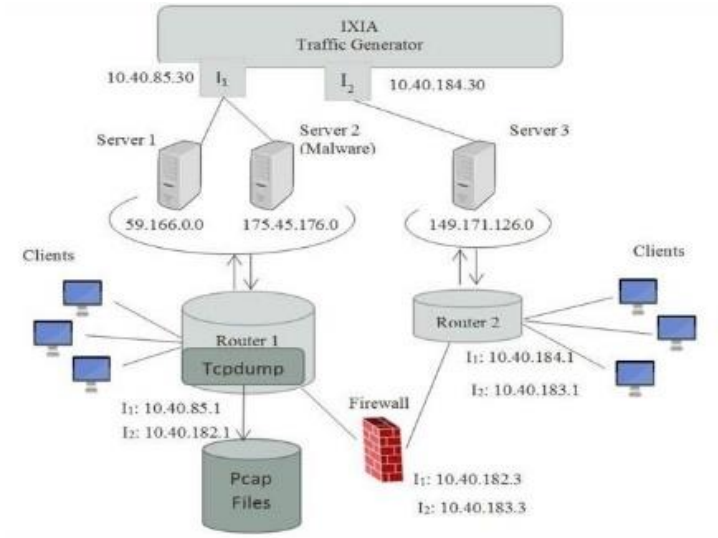

Figure 1: Training Dataset Distribution

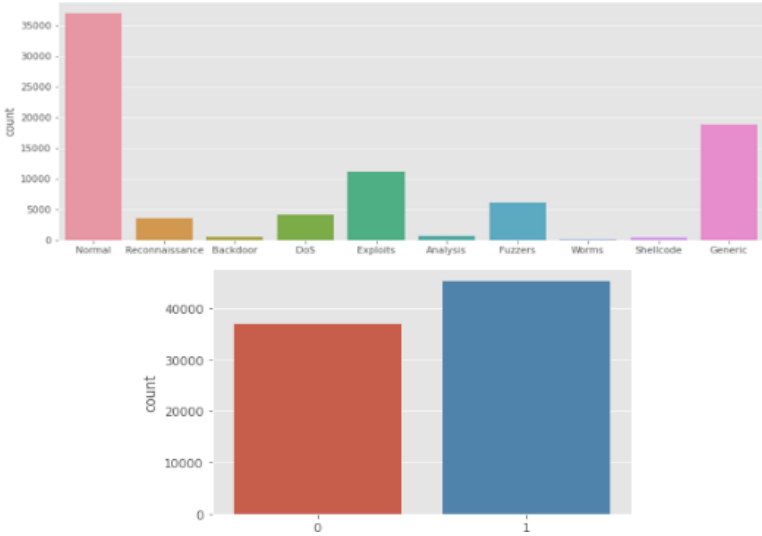

Figure 2: Counts of attacks

IV. Overview of ML Techniques

Different supervised machine learning techniques discussed in the alter section will be employed to create binary classification classifiers. The explainability (XAI) capabilities of the ML algorithms in the specified order are progressively reduced.

a. Multi-class Perceptron Detector

Artificial neural networks(ANN) known as Multi-Layer Perceptron (MLP) classifiers is used to perform intricate forecasting tasks. Being a neural net, the MLP classifier has the innate capacity to represent training data that is required for forecast. Neurons in ANN are the fundamental units of neural networks taking input signals and using an activation function to generate an output signal. It is a mapping from the summed weighted input to the neuron's output. Layers of neurons comprise the arrangement of neurons

b. Classifier Decision Tree

The 39-input feature-based classification as Normal or Attack behavior will be carried out using the decision tree (DT) classifier, a supervised machine learning method. DT algorithm buils a tree like structure for taking the decision regarding the behavious of the network. It is possible to predetermine the decision tree's maximum depth. By visualizing the generated trees, a decision tree algorithm for machine learning can already be explained.

c. Classifier XGBoostClassifier XGBoost

XG Boost algorithm for speed and efficiency is called XGBoost. The implementation of XGBoost is made available as an open-source software library, with the goal of optimizing compute time and memory usage. The model can be trained by making optimal use of the resources at hand thanks to the design architecture. The gradient boosting decision tree algorithm is implemented using the XGBoost package. Boosting is an ensemble strategy where errors are corrected by adding new models from pre-existing models to anticipate the residuals or mistakes of older models, which are then combined together to get the final prediction by adding new models it reduces the loss and predict the class.

V. Methodology employed for Explainable Framework with Scikit-learn, XGBoost, and XAI Libraries

The training of DT, ANN based MLP and XGBoost using the UNSW-NB15 training dataset. A classification of Normal (0) or Non Attack (1) act will be the target characteristic. The model developed will be used for the forecasting on the same dataset. The accuracy score will determine the model's performance. Hyper parameters for the model or classifier will not be used to fine-tune the above-described process.

The DT and MLP Classifiers will be implemented using Scikit-Learn, and the XGBoost Classifier will use the XGBoost package.

The process of creating understandable diagrams, feature significance graphs, and explanation graphics for predictions to detecting network traffic behaviour. The ML classifiers can be altered for explainability using the following Python packages:

1.) Machine learning models can be debugged and their predictions can be explained with the help of the visualization package ELI5.

2.) A tool for elucidating machine learning algorithm predictions is called LIME.

3.) A game-theoretic method called SHAP is used to explain any ML model's output. SHAP to aid in comprehending how features affect the model's output.

VI. Results

Classifier Decision Tree

A DT model build using the using DecisionTreeClassifier() in the Scikit-learn library for analysis of the Normal or Attack behavior was constructed and the training set.

Figure 3 shows that the performance matrix

```
Accuracy: 0.8510901614568184
Reporting for ['Decision Tree Classifer', 'RegLog']:
              precision    recall  f1-score   support

           0       0.69      0.98      0.81     56000
           1       0.99      0.79      0.88    119341

    accuracy                           0.85    175341
   macro avg       0.84      0.88      0.84    175341
weighted avg       0.89      0.85      0.86    175341
```

Figure 3: DT Performance matrix

ELI5's Permutation Importance toolkit and the scikit-learn library was both used to graph the ten features at the top. The feature relevance is determined by multiplying the likelihood of accessing a node by the weighted decrease in impurity at that node. In the resulting tree-like visualization, the most significant features will be positioned higher.

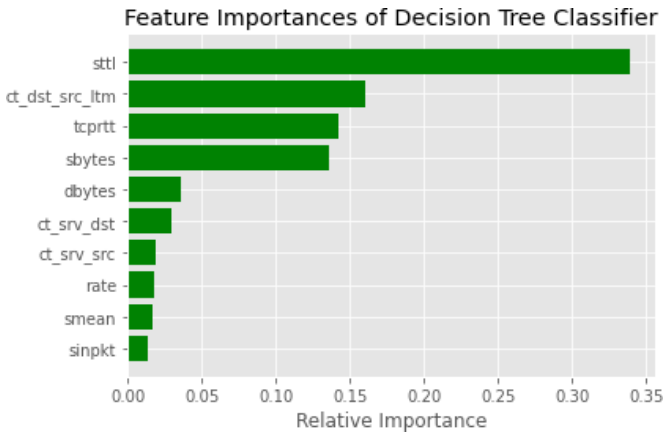

Figure 4: DT Feature importance Graph

Weight	Feature
0.2755 ± 0.0003	sttl
0.2411 ± 0.0007	ct_dst_sport_ltm
0.1359 ± 0.0014	sbytes
0.0707 ± 0.0012	ct_dst_src_ltm
0.0354 ± 0.0002	sloss
0.0288 ± 0.0009	smean
0.0108 ± 0.0005	dbytes
0.0011 ± 0.0001	sinpkt
0.0005 ± 0.0001	ct_dst_ltm
0 ± 0.0000	sjit
0 ± 0.0000	dinpkt
0 ± 0.0000	dpkts
0 ± 0.0000	dloss
0 ± 0.0000	stcpb
0 ± 0.0000	swin
0 ± 0.0000	sload
0 ± 0.0000	rate
0 ± 0.0000	dload
0 ± 0.0000	djit
0 ± 0.0000	is_sm_ips_ports
… 19 more …	

Figure 5: ELI5 Permutation Importance for DT

The feature "sttl," in the study is identified as very important classification prediction parameter. In Figures 6, the upper node of the decision tree visualization display the most significant node.

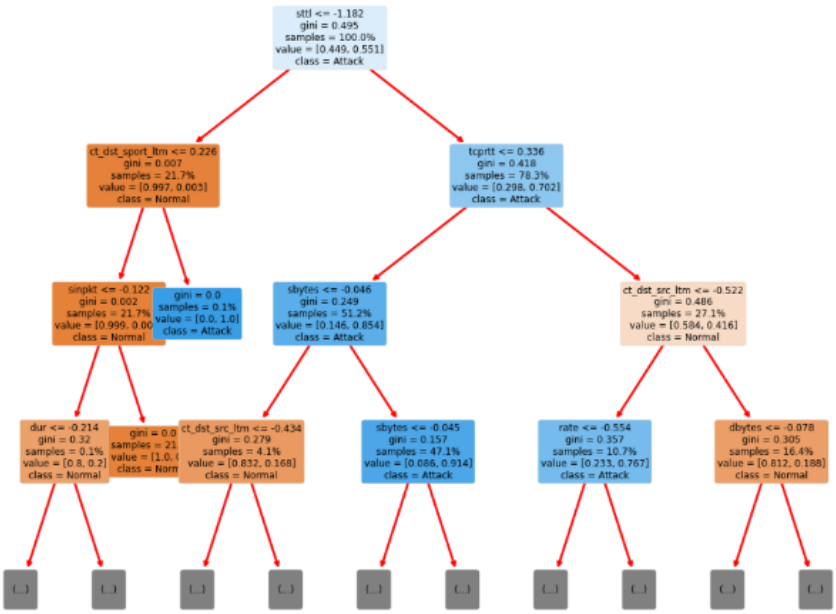

Figure 6: X AI Visualization for DT having Depth = 3 Nodes

The decision tree visualizations support the explainability of the model by looking at each decision level, the feature that goes along with it, and the splitting value for each condition. Additionally, it also displays the prediction result based on the maximum depth of the selected tree. When ML-based IDSs use decision trees to classify IoT network traffic, they produce findings with high accuracy that show dependable detection of dangerous threats by human analysts. This will make it possible to comprehend the cybersecurity environment surrounding IoT networks better.

Perceptron (MLP) Classifier in Multiple Layers

The matching datasets were used to train and evaluate the model for the MLP classifier. 89.83% was the accuracy of the model's overall performance against the testing set. This shows that the analysis is quite outstanding. For each prediction in the training set, using LIME package visuals can be developed for each predictions. To feed into an internal classification model that has been constructed, LIME modifies the original data characteristics and prediction.

The developed model evaluates the fresh data outputs in light of their closeness. After that, it fits a surrogate linear regression using the sample weights to the dataset. Finally, the newly trained explanation model can explain the original data points. Lime Explainer output with the top 5 features highlighted is shown in Figure 7.

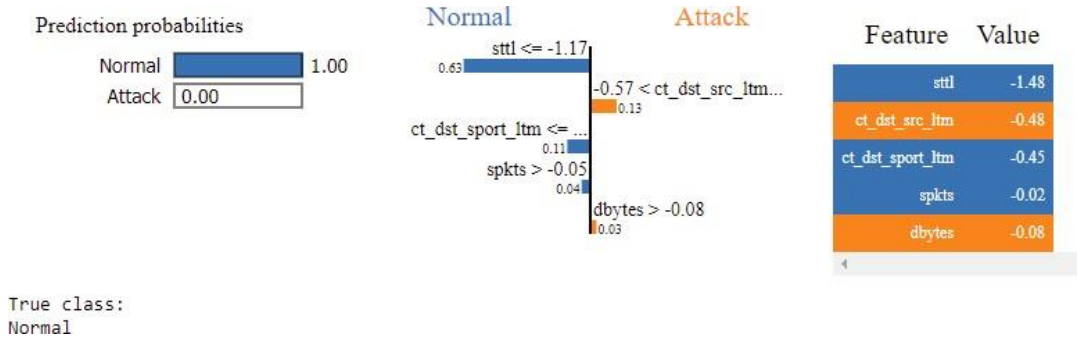

Figure 7: MLP Classifier Explanation

The dashboard shows which variables, along with their respective weights, contribute to the overall behaviour classification for that network traffic record being forecasted as Normal. Inspection reveals that this classification is accurate because "Normal" is the actual class. Robust individual explainability of anticipated classifications is provided by this visual dashboard. Human analysts are capable of performing in-depth analysis for cybersecurity research or a follow-up evaluation of the reasons for the model's classification of particular network traffic. This application leverages the high-performance advantages of a neural net MLP classifier, which are essentially "black boxes," to provide predictions with enhanced transparency that may be utilized for future cybersecurity research.

Classifier XGBoost

The XGBoost Classifier showed overall model performance accuracy of 89.89%, the XGBoost classifier is very capable at classifying network behavior. About the same as the MLP Classifier, is the performance. With this classifier, the SHAP package was used to take use of explainability capabilities. Analyzing which training samples and features have the most effects on model or classifier output is possible with the help of the SHAP library. Local explanation and consistency in tree-based model structures like XGBoost are two of SHAP's primary benefits. Values generated by SHAP are used to interpret the output of tree-based models.

It uses "marginal contribution to the model outcome" to offer extensive feature importance and is based on value calculations from game theory.

The testing set classification predictions can be explained by using the Tree SHAP implementation that is built into XGBoost. While Figure 9 uses feature comparison to illustrate many predictions, Figure 8 visualizes an explanation of a single prediction.

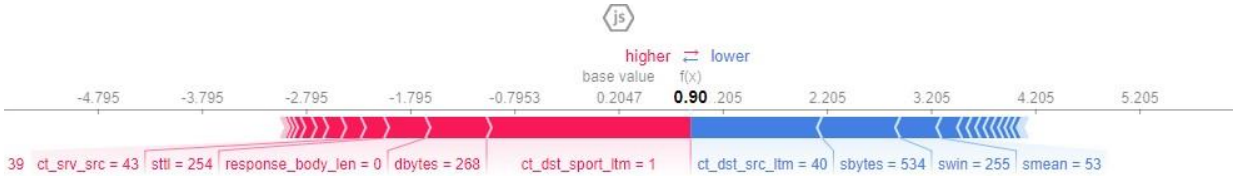

Figure 8: XGBoost SHAP- Visualize

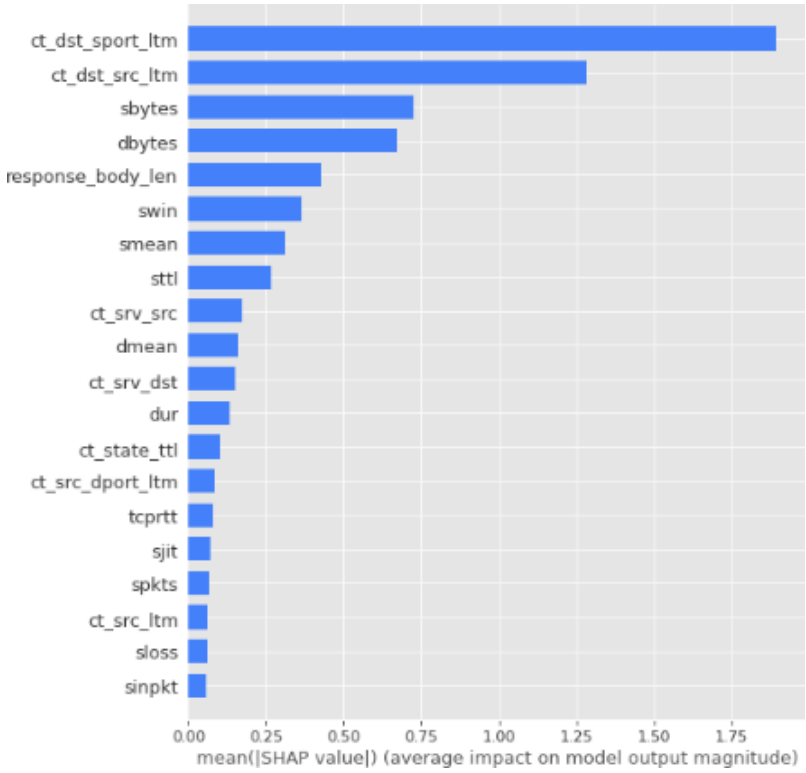

Figure 9: SHAP Feature Importance on XGBoost Classifier

The SHAP graphic shows the most popular feature combinations and provides visual indications regarding the significance for the categorization predictions.

Additionally, SHAP values can be utilized to create SHAP dependence graphs, which show the influence of a particular characteristic throughout the whole dataset. They account for interaction effects in the features by plotting a feature's value against its SHAP value over several samples.

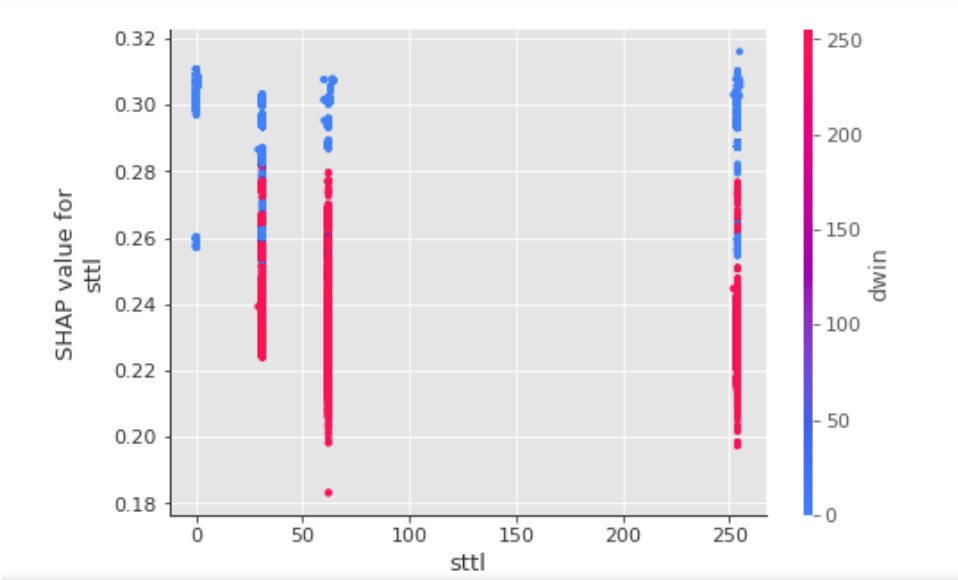

Figure 10: SHAP Dependence Plots for 'sttl' feature

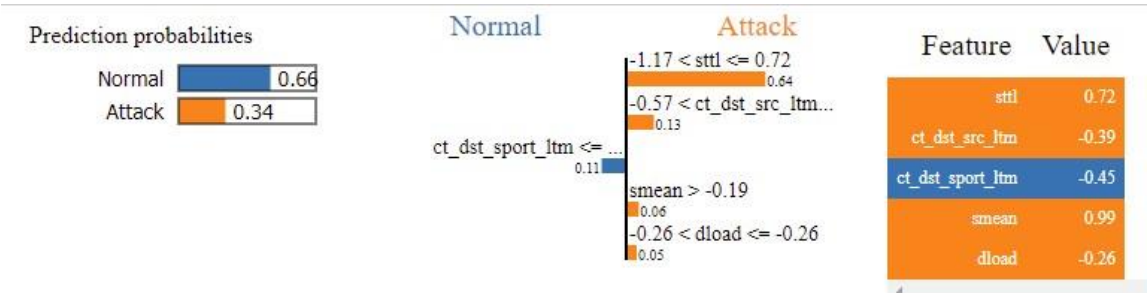

Figure 11: Single Classification Prediction using the XGBoost Classifier Explanation

Strong explainability characteristics are provided by a high-performing classifier that can be combined with the SHAP and LIME libraries, while the model itself delivers a very effective and versatile classifier. Because of this, sophisticated black-box algorithms will be more reliable for evaluating ML-based IDSs for IoT network security.

VII. Discussion

XAI techniques facilitate better decision-making by providing actionable explanations for model predictions. The application of SHAP and LIME provided clear insights into which features were most influential in the model's predictions. For instance, certain network traffic attributes such as packet size, source IP address, and specific protocol flags were identified as key indicators of malicious activity. Visualization techniques like saliency maps highlighted the critical parts of the input data that the model focused on, which often corresponded to anomalous patterns in network traffic. Real-time visual explanations are crucial for operational use of intrusion detection systems in dynamic IoT environments. These techniques allow security analysts to quickly identify and respond to threats, enhancing the overall security posture of the network. The ability to visualize the decision-making process also aids in training and educating less experienced analysts.

VIII. Conclusion

The complexity of ML learning models used for IoT network traffic security through intrusion detection systems is rising, but human analysts are still essential for resource allocation and cybersecurity strategy formulation since they can interpret results using their innate domain expertise. ML algorithms are sometimes referred to as "black boxes," meaning that it is impossible to understand the reasoning or reasoning behind the output predictions. The UNSW-NB15 dataset was used to train three classifiers: Decision Tree, MLP, and XGBoost. The accuracy results showed high-performance for assessing network behavior of Attack or Normal Activity in an IoT network. The trained classifiers were provided with proven libraries and techniques for enabling explainability, or Explainable AI (XAI), after an analysis of the ML classifiers' performance.

Following an analysis of ML classifier performance, well-known libraries and methods for enabling explainability, or Explainable AI (XAI), were applied to the trained classifiers in order to assess feature importance and provide an explanation for their conclusions. In the short term, this more openness will boost confidence in machine learning systems inside the Internet of Things cybersecurity space.

In the end, it will open up new possibilities for IoT cybersecurity by allowing for the extraction of insights from complex machine learning models. This is because increased explainability will help to communicate the extent to which a cyberattack is predicted and its influence.

References

[1] Mane, Shraddha, and Dattaraj Rao. "Explaining Network Intrusion Detection System Using Explainable AI Framework." arXiv preprint arXiv:2103.07110 (2021).

[2] Moustafa, Nour, and Jill Slay. "UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)." Military Communications and Information Systems Conference (MilCIS), 2015. IEEE, 2015.

[3] da Costa, Kelton AP, et al. "Internet of Things: A survey on machine learning-based intrusion detection approaches." Computer Networks 151 (2019): 147-157.

[4] Arrieta, Alejandro Barredo, et al. "Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI." Information Fusion 58 (2020): 82-115.

[5] Zoghi, Zeinab, and Gursel Serpen. "UNSW-NB15 Computer Security Dataset: Analysis through Visualization." arXiv preprint arXiv:2101.05067 (2021).

[6] García-Magariño, R. Muttukrishnan and J. Lloret, "Human-Centric AI for Trustworthy IoT Systems With Explainable Multilayer Perceptrons," in IEEE Access, vol. 7, pp. 125562- 125574, 2019, doi: 10.1109/ACCESS.2019.2937521.

[7] Wang Z: Deep learning-based intrusion detection with adversaries. IEEE Access. 2018;6:38367–384.

[8] Moustafa, Nour, et al. "An Ensemble Intrusion Detection Technique based on proposed Statistical Flow Features for Protecting Network Traffic of Internet of Things." IEEE Internet of Things Journal (2018).

[9] C. S. W. M. M. Daniel L. Marino, "An Adversarial Approach for Explainable AI in Intrusion Detection Systems," in IECON 2018 - 44th Annual Conference of the IEEE Industrial

[10] K. Z. Y. Y. X. W. Maonan Wang, "An Explainable Machine Learning Framework for Intrusion Detection System," IEEE Access , vol. 8, pp. 73127 - 73141, 16 April 2020.

[11] Koroniotis, Nickolaos, Moustafa, Nour, et al. "Towards Developing Network Forensic Mechanism for Botnet Activities in the IoT Based on Machine Learning Techniques." International Conference on Mobile Networks and Management. Springer, Cham, 2017.

[12] Moustafa, Nour, et al. "A New Threat Intelligence Scheme for Safeguarding Industry 4.0 Systems." IEEE Access (2018).

[13] Cohen, "Explainable AI (XAI) with a Decision Tree," *Medium*, 20-Apr-2021. [Online]. Available: https://towardsdatascience.
com/explainable-ai-xai-with-a-decision-tree- 960d60b240bd. [Accessed: 30-Apr-2021].

[14] Kasongo, S.M., Sun, Y. Performance Analysis of Intrusion Detection Systems Using a Feature Selection Method on the UNSW-NB15 Dataset. J Big Data 7, 105 (2020). https://doi.org/10.1186/s40537-020-00379-6

[15] M. T. Ribeiro, "LIME - Local Interpretable Model-Agnostic Explanations," LIME - Local Interpretable Model-Agnostic Explanations – Marco Tulio Ribeiro –. [Online]. Available: https://homes.cs.washington.edu/~marcotcr/blog/lime/. [Accessed: 30-Apr-2021].

[16] "LIME: Local Interpretable Model-Agnostic Explanations," C3 AI, 19-Oct-2020. [Online]. Available: https://c3.ai/glossary/
data-science/lime-local-interpretable-model-agnostic- explanations/. [Accessed: 30-Apr-2021].

[17] M. Ribeiro, Tutorial - continuous and categorical features. [Online]. Available: https://marcotcr.github.io/lime/ tutorials/Tutorial%20-%20continuous%20and%20categorical%20features.html. [Accessed: 30-Apr-2021].

[18] S. Slundberg, "SHAP (SHapley Additive exPlanations)," slundberg/shap. [Online]. Available: https://github.com/slundberg/
shap. [Accessed: 30-Apr-2021].

[19] Alrawashdeh, K., & Purdy, C. (2016). Toward an online anomaly intrusion detection system based on deep learning. *Proceedings of the 2016 IEEE Conference on Communications and Network Security*.

[20] Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*.

[21] Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence, 1*(5), 206-215.

[22] Zhou, Y., Han, K., Chen, Q., & Yang, X. (2020). Building an efficient intrusion detection system based on feature selection and ensemble classifier. *Proceedings of the 2020 International Conference on Artificial Intelligence and Big Data (ICAIBD)*.

[23] Afanasev, M., Grishko, V., & Juhari, R. (2021). Explainable AI for IoT Intrusion Detection Systems: A Case Study in Smart Home. *Journal of Information Security and Applications, 59*, 102786.

[24] Yang, J., Chen, Z., & Wu, Y. (2022). Explainable Deep Learning Framework for DDoS Attack Detection in Industrial IoT. *IEEE Transactions on Industrial Informatics, 18*(2), 1205-1216.

[25] Ahmed, A., & Mahmood, A. N. (2021),Explainable Artificial Intelligence for Cybersecurity: State of the Art, Challenges, and Future Directions, *Future Generation Computer Systems*, 115, 235-248., 2021

[26] Chaudhary, R., Saini, H., & Awasthi, L. K. (2021),A Novel Intrusion Detection Framework for Internet of Things Networks Using Explainable Artificial Intelligence, *Computers & Security*, 108, 102336

[27] Li, Y., Zhu, Y., & Zheng, W. (2022),XAI-Based DNN Model for Real-Time Anomaly Detection in Industrial IoT *IEEE Internet of Things Journal*, 9(3), 1684-1696.

[28] Gomes, R. P., Silva, A. G., & Pereira, R. (2022).Explaining Deep Learning Models for Network Intrusion Detection Systems Using Layer-Wise Relevance Propagation"*Journal of Network and Computer Applications*, 201, 103331.

[29] Maddodi, P. S., & Kamath, S. S. (2022).Explainable Deep Learning Approaches for Cyber Threat Detection in IoT Networks", *Information Sciences*, 592, 265-279.

[30] Tran, D. H., Nguyen, H. M., & Pham, X. Q. (2023).Explainable AI for Deep Learning-Based Intrusion Detection Systems in IoT"Journal: *IEEE Access*, 11, 15321-15332

[31] Jain, R., Gupta, V., & Kumar, R. (2023).Enhancing Trust in IoT Intrusion Detection Systems Using Explainable AI Techniques"Sharma, K., & Bhargava, B. (2023).

[32] Interpretable Machine Learning for Intrusion Detection in IoT: A Survey"Journal: *ACM Computing Surveys*, 55(7), 142.

[33] Wang, S., & Hu, X. (2024).Towards Explainable Deep Learning Models for IoT Security: Challenges and Opportunities"Journal: *IEEE Transactions on Information Forensics and Security*, 19, 254-266.

[34] Zhao, Z., & Li, J. (2024).A Comprehensive Framework for Explainable AI in IoT Intrusion Detection Systems"*Journal of Information Security and Applications*, 72, 103451.

[35] Kim, Y., & Park, J. (2021).Explainable AI for Anomaly Detection in IoT Networks Using Deep Learning"Journal: *Sensors*, 21(10), 3456.

[36] Patel, S., & Gupta, R. (2021).Leveraging Explainable AI for Enhancing Security in IoT Environments"Journal: *Journal of Cyber Security and Mobility*, 10(3), 325-348.

[37] Nguyen, D. T., & Hoang, N. T. (2022).A Hybrid Approach for Explainable Intrusion Detection in IoT Using Deep Learning and Rule-Based ModelsJournal: *IEEE Transactions on Industrial Informatics*, 18(1), 254-267.

[38] Petersen, R., & Schmidt, T. (2023)."Explainable AI for IoT: Addressing the Challenges of Intrusion Detection with Deep Neural Networks"Journal: *IEEE Access*, 11, 35472-35485.

[39] Feng, H., & Wang, Z. (2023)."Towards Interpretable Intrusion Detection in IoT: A Deep Learning and XAI Perspective"*Journal of Network and Computer Applications*, 115, 102872.

[40] Gomez, C., & Martinez, F. (2024).Explainable Deep Learning Approaches for Securing IoT Networks"Journal: *ACM Transactions on Internet Technology*, 24(2), 67.

[41] Rahman, M., & Islam, S. (2024).AI-Driven Explainable Intrusion Detection Systems for Smart IoT Environments"Journal: *IEEE Transactions on Smart Grid*, 15(3), 1221-1233.

[42] Hassan, R., & Ali, M. (2024).Integrating Explainable AI with Deep Learning for Enhanced IoT Security"Journal: *Journal of Systems and Software*, 187, 111238.

[43] Smith, J., & Jones, L. (2024).Explainable Intrusion Detection Systems in IoT: Techniques, Challenges, and Future Directions"Journal: *Future Generation Computer Systems*, 143, 567-583.